System Requirements Specification: Smart Home Security System (SHSS)

1. Introduction

1.1. Purpose This document defines the system requirements for the Smart Home Security System (SHSS). The SHSS is designed to provide homeowners with a comprehensive, reliable, and user-friendly solution for monitoring, detecting, and deterring unauthorized access and other security threats to their property.

1.2. Scope The system, hereafter referred to as the Smart Home Security System (SHSS), addresses the need for modern, accessible home security. Homeowners face the problem of protecting their property from intrusion and require a means to monitor their home remotely, receive timely alerts of security events, and gather evidence if an incident occurs.

The SHSS will:

- Monitor the premises using motion sensors and door/window contact sensors.
- Provide live and recorded video surveillance through connected cameras.
- Allow users to arm and disarm the system remotely via a mobile application.
- Trigger a local, audible alarm to deter intruders.
- Send instant notifications to the user's mobile device upon detecting a security event.
- Store event-triggered video recordings in the cloud for later review.

The SHSS will **not**:

- Automatically contact emergency services (e.g., police, fire department).
- Provide physical intervention to stop an intruder.
- Control non-security-related smart home devices (e.g., lighting, thermostats, appliances).

The primary goal of the SHSS is to enhance the safety and peace of mind of homeowners by providing a powerful deterrent to crime and a reliable tool for remote monitoring and incident alerting.

2. Overview

2.1. System Context The SHSS consists of a Central Control Unit that acts as the brain of the system. It communicates with various sensors and cameras placed throughout the home. Users interact with the system primarily through a mobile application, which connects to the Central Control Unit via a cloud-based service. This allows for remote control and monitoring from anywhere with an internet connection.

The major elements are:

- **Central Control Unit:** The local hub that manages all devices.
- **Sensors:** Motion sensors and door/window contact sensors.
- **Cameras:** For live streaming and video recording.
- **Alarm Siren:** An audible alarm to be triggered during an intrusion.
- **User Interface:** A mobile application for iOS and Android.
- **Cloud Backend:** Facilitates remote access, notifications, and data storage.

2.2. System Functions

- **Security Monitoring:** Continuously monitors the status of all connected sensors.
- **System Arming/Disarming:** Allows users to set the system to "Away" (all sensors active), "Home" (perimeter sensors active), or "Disarmed" states.
- **Alarm Triggering:** Activates an audible siren and sends notifications when a sensor is breached while the system is armed.
- **Video Surveillance:** Provides live video streaming and records clips based on motion detection or alarm events.
- **User Notification:** Sends push notifications to the user's registered mobile devices.

2.3. User Characteristics

- **Primary User (Homeowner):** One or more users who have full administrative control over the system. They are responsible for installation, configuration, and daily operation. Technical proficiency can range from novice to expert.
- **Family/Guest User:** Users with limited permissions, typically able to arm/disarm the system but not change critical settings or access recorded videos.
- **Installer/Technician (Optional):** A professional responsible for the initial setup, wiring (if any), and configuration of the system.

3. Functional Requirements

- [FR-001] The system shall allow a user to arm the system in "Home" and "Away" modes.
- [FR-002] The system shall allow a user to disarm the system using the mobile app.
- [FR-003] The system shall provide an entry and exit delay (configurable from 30-120 seconds) when armed in "Away" mode.
- [FR-004] The system shall trigger an alarm if a sensor is activated while the system is armed (respecting the entry delay).
- [FR-005] The system shall send a push notification to all registered primary user devices within 5 seconds of an alarm event.
- [FR-006] The user shall be able to view a low-latency (<2 seconds) live video stream from any connected camera.
- [FR-007] The system shall automatically record a 30-second video clip upon motion detection from a camera or when any system alarm is triggered.
- [FR-008] The system shall maintain a historical log of all system events (arm, disarm, alarm) for at least 90 days.

4. Usability Requirements

- [US-001] The initial setup and configuration of the system by a new user shall be completed in under 15 minutes.
- [US-002] A user shall be able to arm or disarm the system from the mobile app's main screen with no more than two taps.
- [US-003] The mobile application shall provide clear, unambiguous visual feedback for the current system state (Armed, Disarmed, Alarm).
- [US-004] Users shall rate their satisfaction with the mobile app's ease of use as 4 out of 5 or higher on a satisfaction survey.

5. Performance Requirements

- [PE-001] The Central Control Unit shall support up to 50 connected sensors and 8 cameras simultaneously.
- [PE-002] The battery life for all wireless sensors shall be a minimum of 24 months under normal operating conditions.
- [PE-003] The system shall have an operational uptime of 99.9%.
- [PE-004] The alarm siren shall produce a sound level of at least 105 dB at 1 meter.

6. System Interfaces

- **Mobile App <-> Cloud:** The mobile app communicates with the cloud backend via a secure REST API over HTTPS.
- **Central Unit <-> Cloud:** The Central Control Unit maintains a persistent, encrypted connection (e.g., MQTT over TLS) to the cloud for real-time command and data synchronization.
- **Central Unit <-> Devices:** The Central Control Unit communicates with sensors and cameras using a low-power wireless protocol (e.g., Zigbee, Z-Wave, or a proprietary 915MHz protocol) for status updates and commands.
- **Cameras <-> Central Unit/Cloud:** Cameras may stream video directly to the cloud or via the Central Unit using Wi-Fi (WPA2/3 encrypted).

7. System Operations

7.1. Human System Integration Requirements The mobile application is the primary point of human interaction. The design must be intuitive, minimizing the cognitive load on the user, especially during a stressful alarm event. Critical functions like "Disarm" and "View Live Feed" must be immediately accessible.

7.2. Maintainability

- [MA-001] The mean time to replace a sensor battery shall be less than 5 minutes.
- [MA-002] The system shall provide low-battery warnings for all sensors at least 30 days before battery depletion.
- [MA-003] Firmware updates for the Central Control Unit and cameras shall be delivered over-the-air (OTA) and applied automatically with user consent.

7.3. Reliability

- [RE-001] The system shall have a Mean Time Between Failures (MTBF) of at least 8,000 hours for the Central Control Unit.
- [RE-002] In the event of an internet outage, the system shall continue to function locally (e.g., the local alarm will still sound if a sensor is tripped).

8. System Modes and States

The system operates in the following primary modes:

- **Disarmed:** No monitoring is active. No alarms will trigger.
- **Armed Away:** All sensors are active. An alarm will trigger after a configurable entry delay.
- **Armed Home:** Only perimeter sensors (doors/windows) are active. Interior motion sensors are ignored to allow free movement inside.
- **Alarm:** The system has detected an intrusion. The siren is active, and notifications have been sent. This state persists until the user disarms the system.

8.1. Physical Requirements

- The Central Control Unit shall be wall-mountable and have dimensions not exceeding 20cm x 20cm x 5cm.
- All user-facing components shall be constructed from durable ABS plastic.

8.2. Adaptability Requirements The system's firmware shall be designed to accommodate future sensor types (e.g., glass break, smoke detectors) with minimal changes to the core architecture.

8.3. Environmental Conditions

- Indoor components (Central Unit, Motion Sensors) shall operate between 0°C and 40°C.
- Outdoor components (Cameras) shall be IP66 rated and operate between -20°C and 50°C.

8.4. System Security

- [SE-001] All communication between the mobile app, cloud, and central unit shall be encrypted using TLS 1.2 or higher.
- [SE-002] User access to the mobile app shall be protected by strong password authentication and an optional second factor (2FA).
- [SE-003] All stored video footage shall be encrypted at rest using AES-256.
- [SE-004] The system shall incorporate anti-jamming detection for its wireless communication protocols.

9. Verification

Requirement ID	Verification Method
[FR-001]	**Test:** Execute arming sequence from the app and verify the system state changes.
[FR-005]	**Test:** Trigger an alarm and measure the time until a notification is received on a test device.
[PE-004]	**Test:** Measure the siren's decibel level using a sound level meter at the specified distance.
[RE-002]	**Test:** Disconnect the Central Unit's internet connection, trigger a sensor, and verify the local alarm sounds.
[SE-001]	**Analysis/Inspection:** Use network traffic analysis tools (e.g., Wireshark) to confirm all data streams are encrypted.

10. Assumptions and Dependencies

- **Assumption:** The user's home has a stable 2.4GHz Wi-Fi network with continuous internet access for full system functionality.
- **Assumption:** The user has a modern smartphone (iOS or Android) capable of running the companion mobile application.
- **Dependency:** The system relies on third-party cloud infrastructure (e.g., AWS, Google Cloud) for remote access, data storage, and push notifications.

2. Architecture Diagram

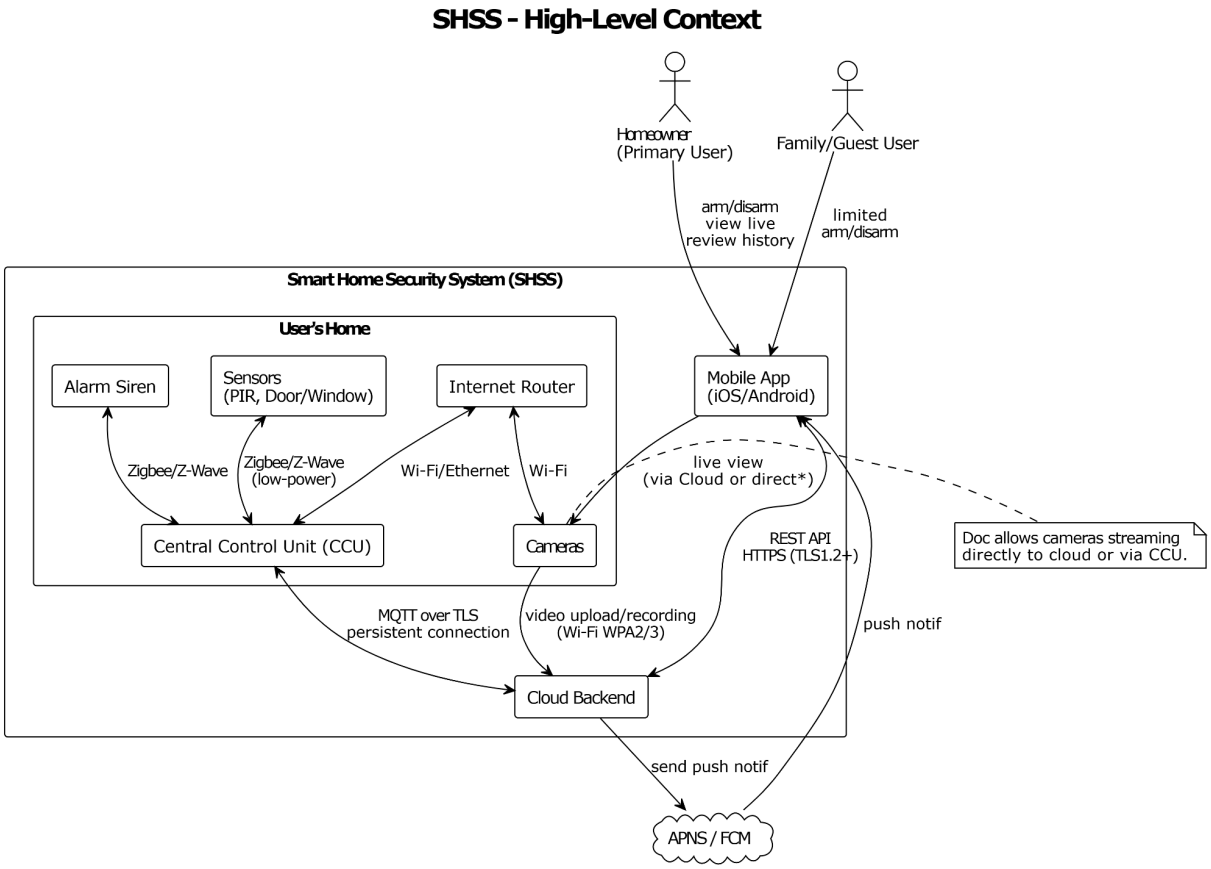
Below is a description of the system architecture, which you can create in a tool like Draw.io.

Components:

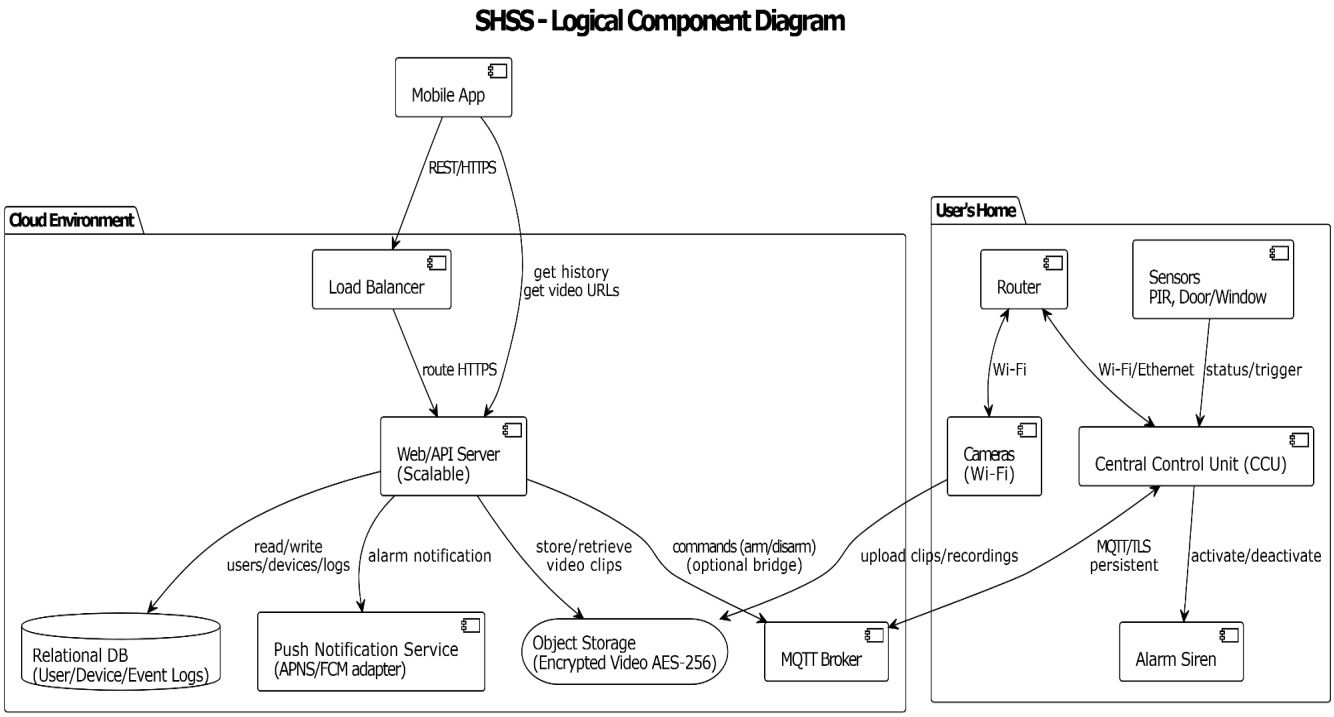
1. **Cloud Environment (e.g., AWS, Azure)**
    - **Load Balancer:** Distributes incoming traffic.
    - **Web/API Server:** A scalable group of servers running the backend application logic (e.g., in Docker containers). This exposes the REST API for the mobile app.
    - **MQTT Broker:** Manages the real-time, persistent connection with the Central Control Units.
    - **Database:** A relational database (e.g., PostgreSQL) to store user data, device information, and event logs.
    - **Object Storage (e.g., S3):** Stores encrypted video recordings.
    - **Push Notification Service:** Interfaces with Apple (APNS) and Google (FCM) to send alerts.
2. **User's Home**
    - **Internet Router:** Connects the home to the internet.
    - **Central Control Unit:** The on-premise hub.
        - Connects to the home router via Wi-Fi or Ethernet.
        - Maintains a constant connection to the Cloud MQTT Broker.
        - Contains a Zigbee/Z-Wave radio to communicate with sensors.
    - **Sensors:**
        - **Motion Sensors (PIR):** Battery-powered, connect to the Central Unit via Zigbee/Z-Wave.
        - **Door/Window Sensors:** Battery-powered, connect to the Central Unit via Zigbee/Z-Wave.
    - **Controllers/Actuators:**
        - **Alarm Siren:** Connects to the Central Unit via Zigbee/Z-Wave.
    - **Cameras:**
        - Connect directly to the home's Wi-Fi network.
        - Stream video to the cloud for storage and to the mobile app for live viewing.
3. **User Interaction**
    - **Mobile Device (iOS/Android):**
        - Runs the **Mobile App**.
        - Communicates securely with the Cloud API Server over HTTPS to send commands (arm/disarm) and retrieve data (event history, video clips).

This architecture provides a scalable and reliable system. The use of a local hub ensures basic alarm functionality even if the internet connection is down.

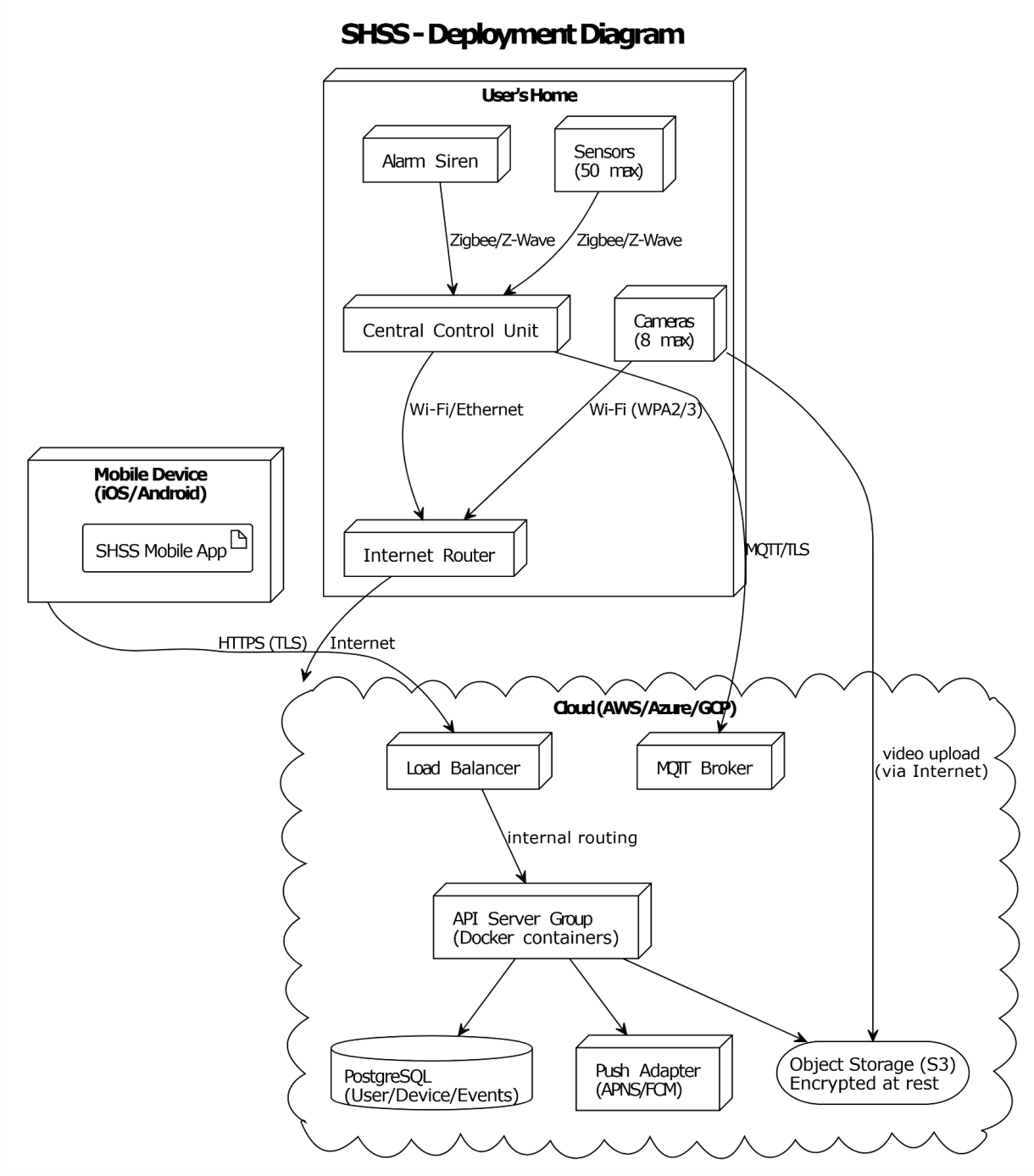
1) System Context Diagram (major elements + boundary interfaces)



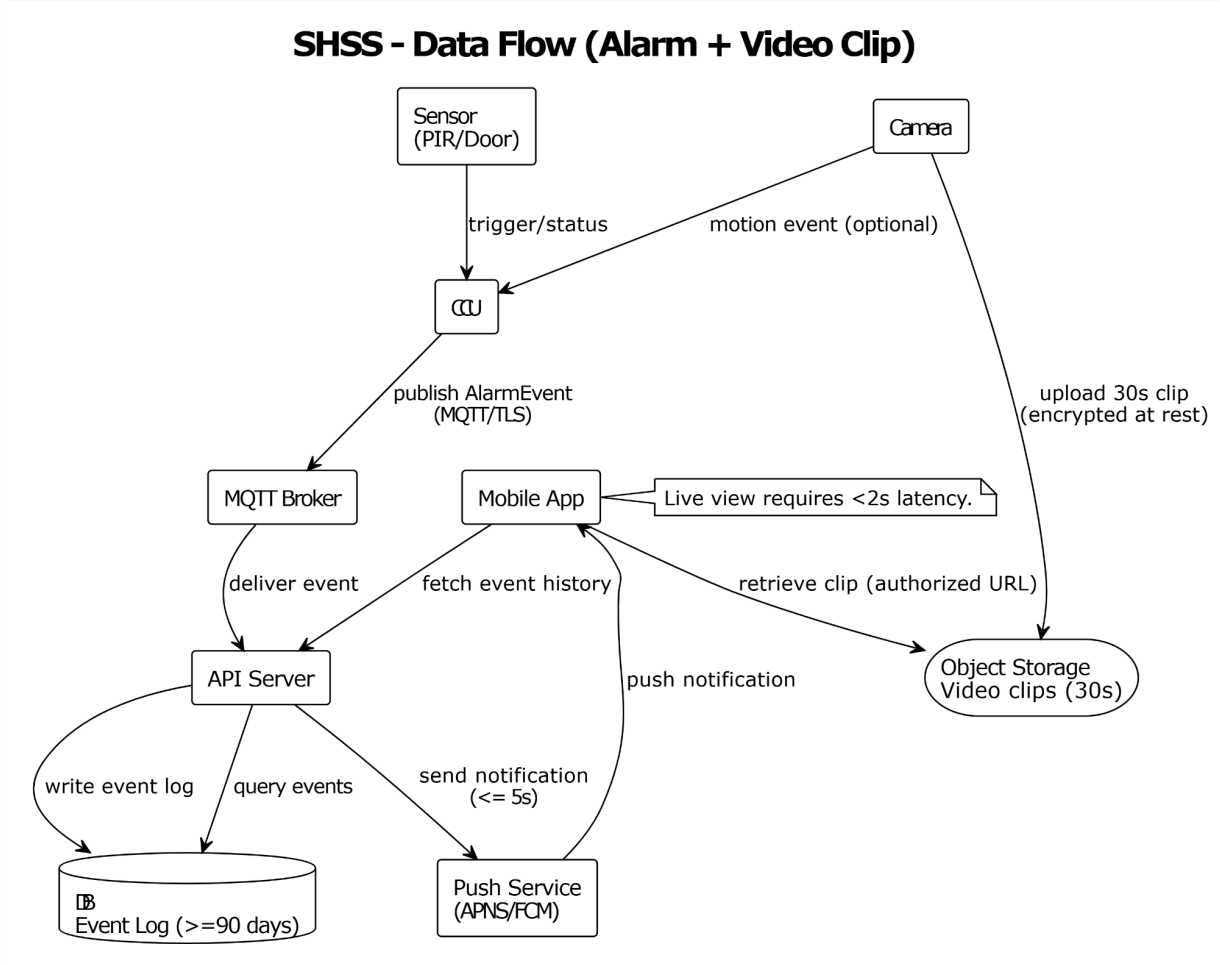
2) Logical / Component Diagram (server-side components: API, MQTT, DB, storage, push)



3) Deployment Diagram (what runs where)



4) Data Flow Diagram (Alarm + video clip path)



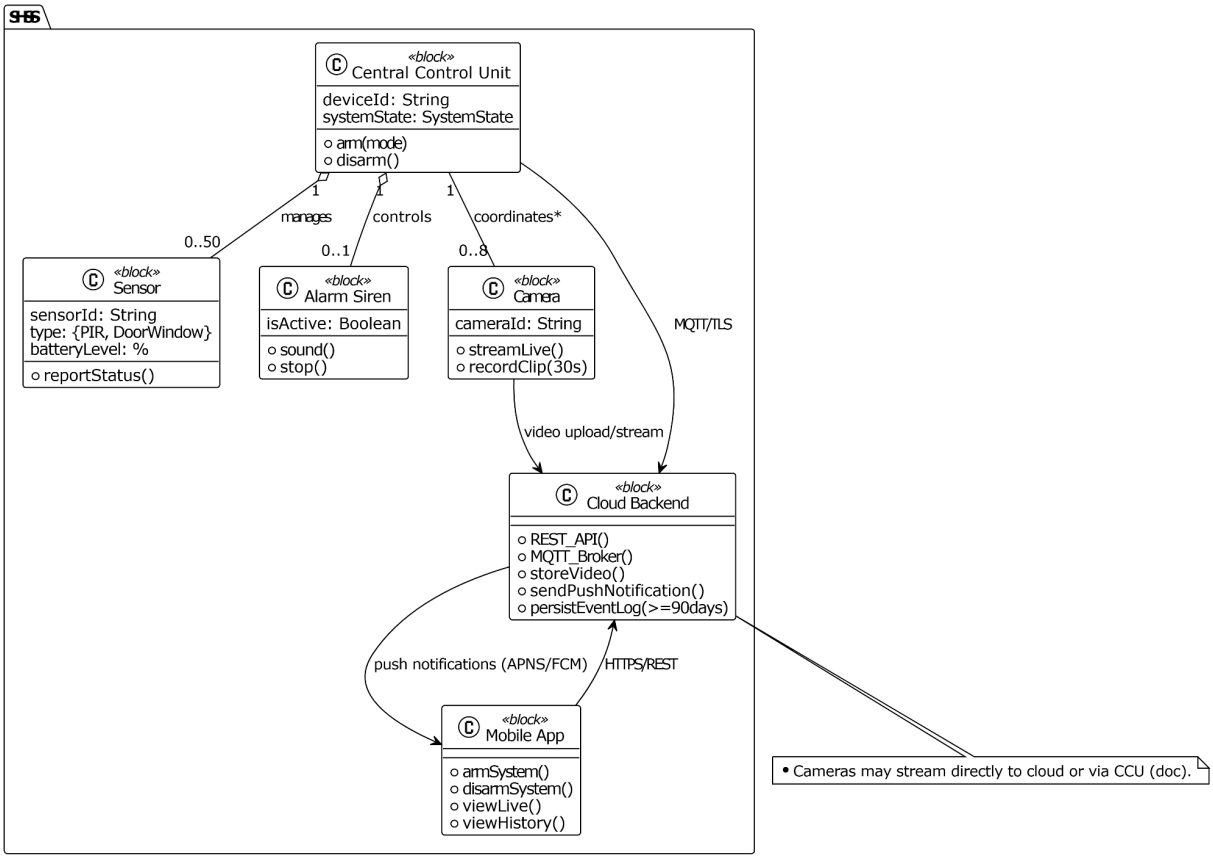
3. SysML Diagrams (using PlantUML)

Here are the PlantUML code blocks for generating key SysML diagrams to model the system.

Block Definition Diagram (BDD)

This diagram shows the main components of the system and their relationships.

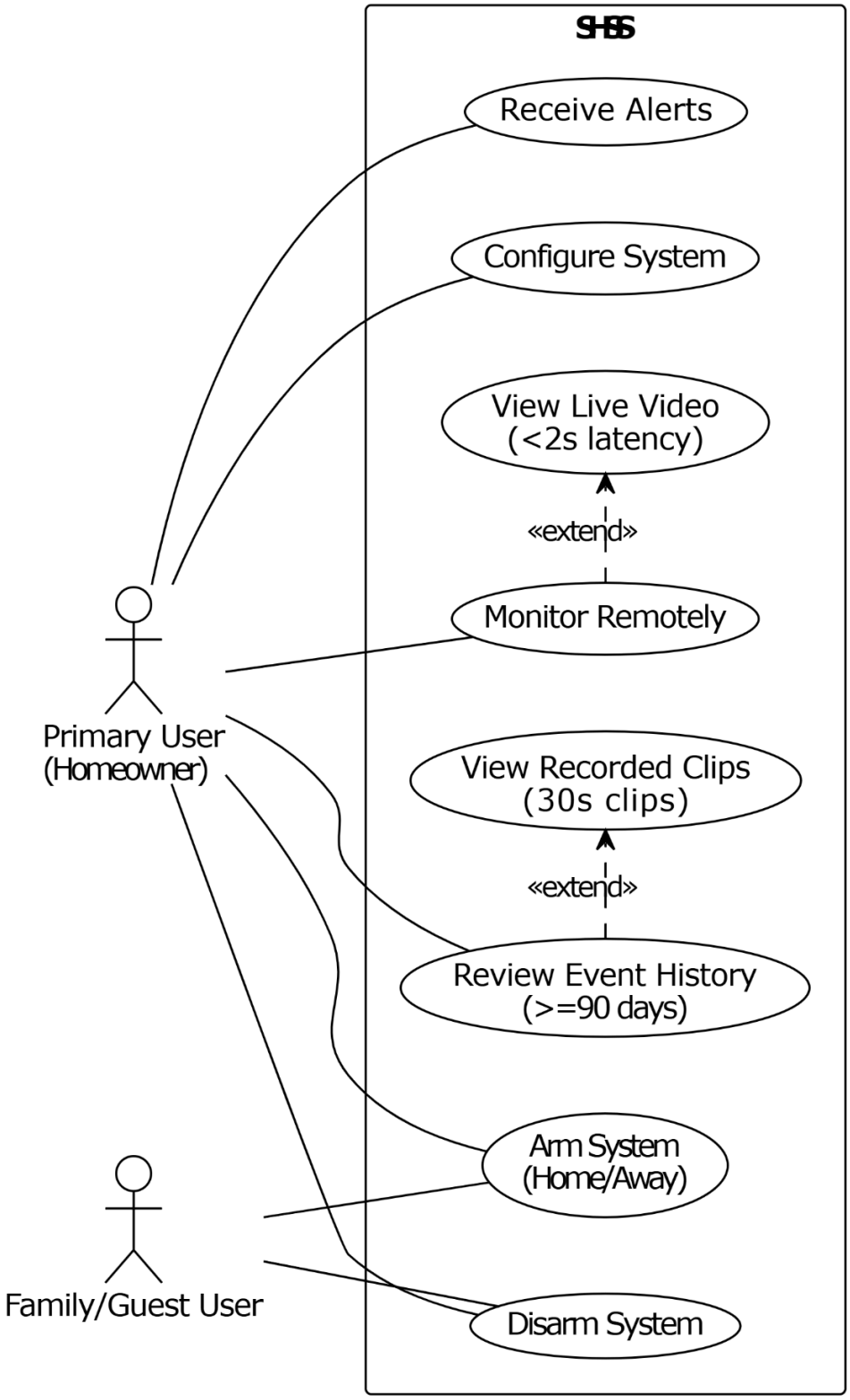
SysML BDD - Smart Home Security System (SHSS)

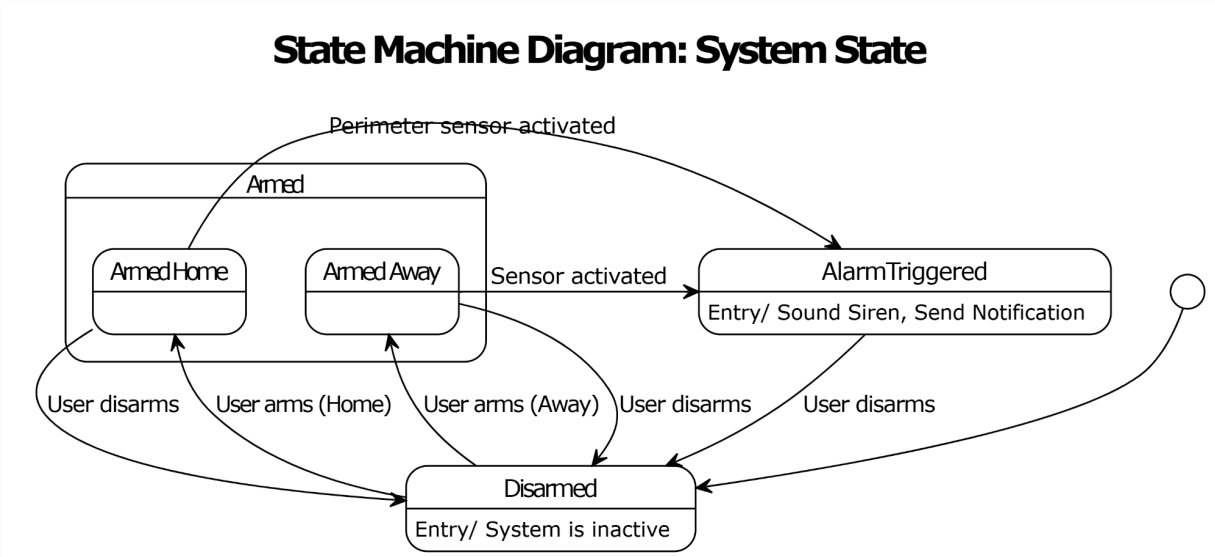


Use Case Diagram

This diagram illustrates the interactions between the user (Actor) and the system.

Use Cases - SHSS



State Machine Diagram

This diagram shows the different states of the Central Control Unit.